Pay to Quit

Robust Team Incentives with Hidden Need

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Team project with hidden need

A project consists of interdependent parts that differ in their *initial quality*.

Each expert privately inspects the initial quality of the part:

- Some parts are already *satisfactory*: no need to repair.
- Some parts are *faulty* and require repair.

Each expert privately chooses whether

- to make a *costly effort* to repair
- or to *leave* it as it is.

For example: a software update release, a medical team, a consulting team, and remedial repairs.

What does the manager do?

A natural contract: pay only experts who *undertake* the repair task.

- Experts with satisfactory parts also want the bonus.
- Undertaking the task does not distinguish productive repair from unnecessary presence.

What is an optimal way to induce high-quality parts in such a team project?

This paper

Setting. Team production with complementarities, hidden need, and unobservable effort

Objective. Robustly induce high quality in every part at the minimum cost.

Solution concept. Belief-free rationalizability (Bergemann & Morris, 2017)

Finding. The optimal contract is pay-to-quit.

Today. Focus on fault-repair technology

Model with symmetric $n = 2$

Team project

A project is managed by two experts, $i \in \{1, 2\}$.

Each Expert i:

- Task type: initial part quality θ_i is faulty L or satisfactory H, with $\Pr(\theta_i = H) = p \in (0, 1)$.
- Effort: chooses $e_i \in \{W, S\}$, where W denotes repair at cost c and S denotes no effort.

Part quality

$$q_i = q(\theta_i, e_i) = \begin{cases} q_H & \text{if } \theta_i = H \\ q_H & \text{if } \theta_i = L \text{ and } e_i = W \\ q_L & \text{if } \theta_i = L \text{ and } e_i = S \end{cases}$$

...Fault-repair technology

Project success

$$P(\{1, 2\}) = 1, \qquad P(\{1\}) = P(\{2\}) = \tfrac{1}{2}, \qquad P(\varnothing) = \tfrac{1}{4}$$

Individual part qualities are complementary.

Team project

1. Each expert privately observes the task type θ_i.
2. Then, chooses whether to undertake the task ($a_i = 1$) or opt out ($a_i = 0$).
   - $a_i = 1$: Experts undertaking the task privately choose effort $e_i \in \{W, S\}$.
   - $a_i = 0$: Opting out is effectively $e_i = S$.

   $\rightarrow$ Actions: $(a_i, e_i) \in \{(0, S), (1, W), (1, S)\}$
3. The project succeeds with $P(Q)$, where Q is the exact set of parts with q_H.
   Everyone observes the project outcome and task undertaking profile $\mathbf{a} = (a_1, a_2)$.

We analyze a simultaneous-move game, where the undertaking profile $\mathbf{a} = (a_1, a_2)$ is observed ex post.

Key feature: opting out reveals no effort $\Rightarrow$ Undertaking choice partially verifies effort choice.

Principal's problem

The manager aims to achieve all-high-quality ($q_1 = q_2 = q_H$) in every outcome at minimum cost.

Contract: Bonus $b_i(\mathbf{a}) \geq 0$, conditioned on the undertaking profile and the project outcome

Payoff and cost: Let $Q = \{j : q_j = q_H\}$.

Expert i's payoff given a realization $(\boldsymbol{\theta}, \mathbf{a}, \mathbf{e})$:

$$u_i = P(Q)\, b_i(\mathbf{a}) \; - \; c\, \mathbf{1}\{e_i = W\}$$

Principal's expected cost under $\mathbf{p}$:

$$C_{\mathbf{p}}(\mathbf{b}) = \mathbb{E}_{\mathbf{p}} \left[P(Q) \cdot \sum_{i \in N} b_i(\mathbf{a}) \right]$$

Remark. Team project

In a team project:

1. An expert with a faulty part faces uncertainty about the other part's quality:
   - q_H: already satisfactory or faulty but repaired.
   - q_L: faulty and unrepaired.
2. Because parts' qualities are complementary, the same contract may support:
   - Repair under optimistic beliefs about the other parts.
   - No repair under pessimistic beliefs.
3. In an organization, information is dispersed across experts, and their beliefs about other parts may differ.

Robust implementation

Objective: every part reaches high quality in every rationalizable outcome.

Belief-Free Rationalizability (Bergemann & Morris, 2017) Def.
For each expert of each type, iteratively eliminate actions that are never best replies to any arbitrary belief over others' surviving types and actions.

This also rules out BNE with low-quality parts under any common prior.

θ_i \ (a_i, e_i)	Opt out $(0, S)$	Undertake & Shirk $(1, S)$	Undertake & Work $(1, W)$
Initial high quality H	q_H	q_H	q_H
Initial low quality L	q_L	q_L	q_H

Two contract forms

Two contract forms can be offered to each expert.

- Participation-Only: $b_i(0, a_{-i}) = 0$. An expert who opts out is not paid.
- Pay-to-Quit: $b_i(0, a_{-i}) > 0$. Opting out is also paid.

We compare (1) PO for both experts with (2) PtQ for both experts.

Benchmark: Participation-Only

PO contracts

Under PO, all types undertake the repair task.

For type H, $e_i = W$ does not improve the quality but incurs cost c.

It remains to incentivize all L-type experts to choose $e_i = W$.

Resolve experts in order (Divide-and-conquer, Segal 2003, Winter 2004).

Expert 1 with $\theta_1 = L$ chooses $e_1 = W$ if

$$b_1 > \frac{c}{P(Q_{-1} \cup \{1\}) - P(Q_{-1})}.$$

Under the worst-case belief $Q_{-1} = \varnothing$, we need $b_1 > b_1^* = 4c$.

Expert 1

	$(0, S)$	$(1, S)$	$(1, W)$
H	q_H	q_H	q_H
L	q_L	q_L	q_H

Expert 2

	$(0, S)$	$(1, S)$	$(1, W)$
H	q_H	q_H	q_H
L	q_L	q_L	q_H

PO contracts

Under PO, all types join the remediation task.

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Resolve experts in order (Divide-and-conquer, Segal 2003, Winter 2004).

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Under the worst-case belief $Q_{-1} = \varnothing$, we need $b_1 > b_1^* = 4c$.

Given that, eliminate $(1, S)$.

Expert 2

	$(0,S)$	$(1,S)$	$(1,W)$
H	q_H	q_H	q_H
L	q_L	q_L	q_H

Expert 2 with $\theta_2 = L$: 2's worst-case belief is $Q_{-2} = \{1\}$.

$$b_2 > \frac{c}{P(\{1,2\}) - P(\{1\})},$$

which requires $b_2 > b_2^* = 2c$.

PO remarks

1. Expected cost: $C^{PO} = P(\{1,2\}) \cdot (4+2)c = 6c$.
2. Everyone undertakes repair tasks regardless of type.
   Undertaking the task reveals neither hidden need nor productive effort.
3. Even if $\boldsymbol{\theta} = \mathbf{H}$ is realized, robust contracts treat experts as if $\boldsymbol{\theta} = \mathbf{L}$.

Can pay-to-quit make undertaking profiles informative about quality?

- If it could, we could reduce the size of quality uncertainty itself.
- It does and is an optimal contract!

Optimal contract: PtQ contract

Pay-to-quit contracts

Suppose $b_i(0, a_{-i}) > b_i(1, a_{-i})$.

For either type:

- $(0, S)$ and $(1, S)$ yield the same quality.
- $(0, S)$ pays a higher success bonus.

$\implies$ $(0, S)$ strictly dominates $(1, S)$ for both experts.

For type H:

- $(0, S)$ and $(1, W)$ yield the same quality.
- $(1, W)$ pays less and incurs the effort cost c.

$\implies$ $(0, S)$ strictly dominates $(1, W)$ for type H.

It remains to induce type-L experts to choose $(1, W)$.

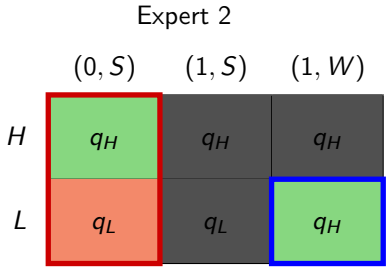
Expert 1

	$(0, S)$	$(1, S)$	$(1, W)$
H	q_H	q_H	q_H
L	q_L	q_L	q_H

Expert 2

	$(0, S)$	$(1, S)$	$(1, W)$
H	q_H	q_H	q_H
L	q_L	q_L	q_H

Undertaking assures quality



PtQ contracts

Expert 1 with $\theta_1 = L$

- Contingencies with $a_2 = 0$: θ_2 can be H or L.
  $\implies b_1^*(a_2 = 0) = 4c$.
- Contingency with $a_2 = 1$: Expert 2 is guaranteed to produce q_H.
  $\implies b_1^*(a_2 = 1) = 2c$.

A small perturbation $\varepsilon_1(a_1 = 0) > \varepsilon_1(a_1 = 1) > 0$ eliminates $(0, S)$.

Expert 1

	$(0, S)$	$(1, S)$	$(1, W)$
H	q_H	q_H	q_H
L	q_L	q_L	q_H

Expert 2

	$(0, S)$	$(1, S)$	$(1, W)$
H	q_H	q_H	q_H
L	q_L	q_L	q_H

PtQ contracts

Expert 1 with $\theta_1 = L$

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  $\implies b_1^*(a_2 = 1) = 2c$.

A small perturbation $\varepsilon_1(a_1 = 0) > \varepsilon_1(a_1 = 1) > 0$ eliminates $(0, S)$.

Expert 2 with $\theta_2 = L$

$b_2^* = 2c$ in any contingency.

Expert 1

	$(0, S)$	$(1, S)$	$(1, W)$
H	q_H	q_H	q_H
L	q_L	q_L	q_H

Expert 2

	$(0, S)$	$(1, S)$	$(1, W)$
H	q_H	q_H	q_H
L	q_L	q_L	q_H

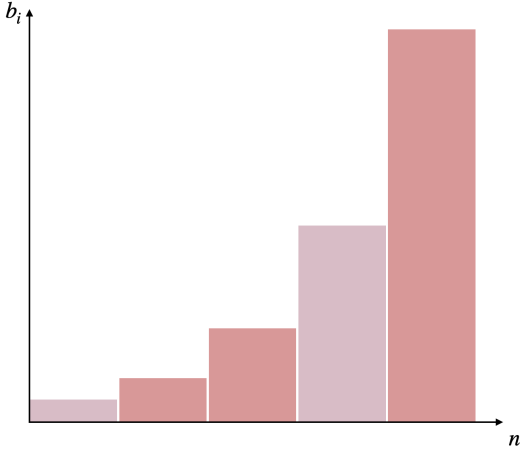
PtQ remarks

1. Expected cost:

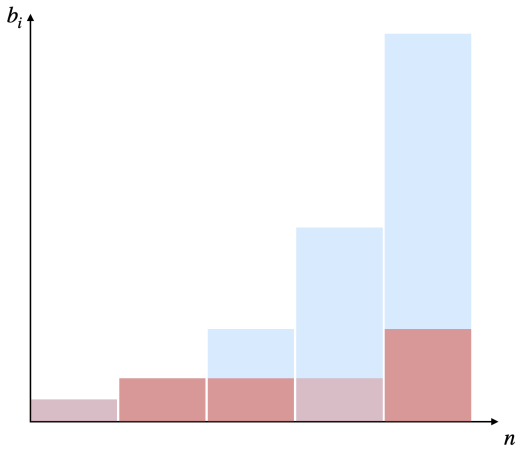
$$C^{PtQ}(p) = P(\{1,2\}) \Big[\underbrace{(1-p)\cdot 2c + p\cdot 4c}_{\mathbb{E}[b_1^*]} + \underbrace{2c}_{b_2^*} \Big] = (4+2p)\,c < 6c = C^{PO}$$

2. Expected cost depends on the prior **p**, but implementation does not.
3. Each expert's quality uncertainty = # later-ranked experts who opt out.

PO vs. PtQ: When 3/5 need fixing



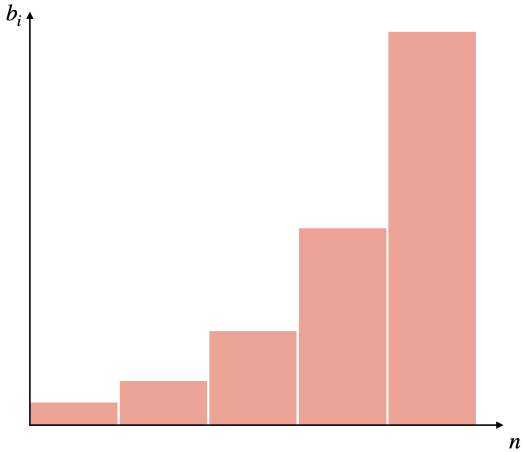
(a) PO



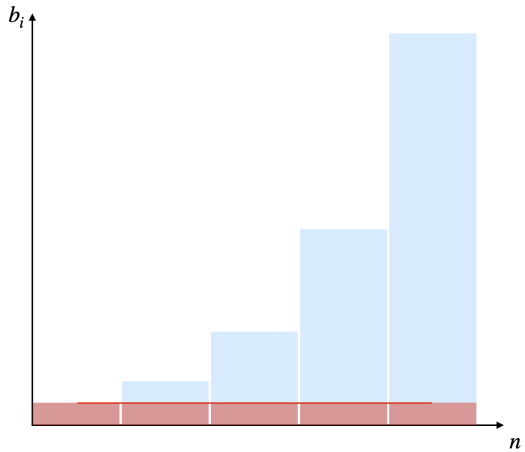
(b) PtQ

In PO, 5 experts undertake and receive bonuses sized with a full hierarchy of length 5.
In PtQ, 3 undertake, and the hierarchy reduces to $3 = (5 - 3 + 1)$.

PO vs. PtQ: When 5/5 need fixing



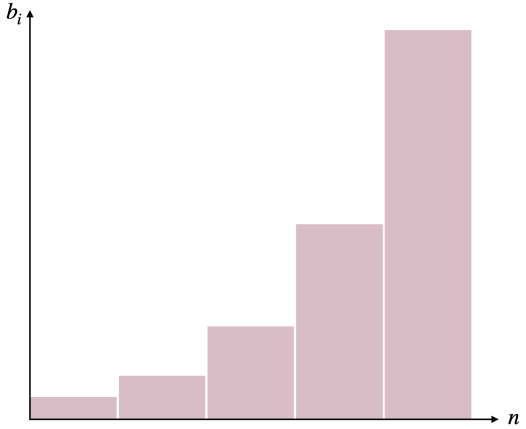
(a) PO



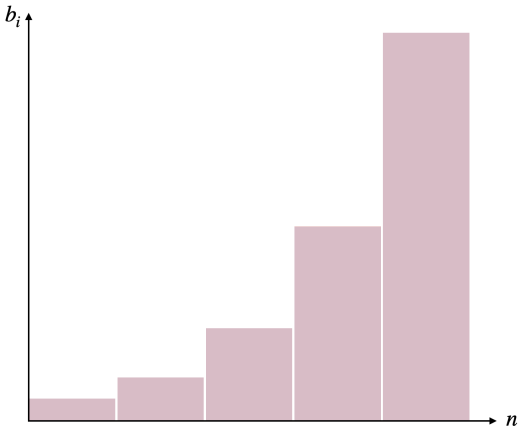
(b) PtQ

In PO, 5 experts undertake and receive bonuses sized with a full hierarchy of length 5.
In PtQ, 5 undertake, and the hierarchy reduces to $1 = (5 - 5 + 1)$.

PO vs. PtQ: When 0/5 need fixing



(a) PO



(b) PtQ

In PO, 5 experts undertake and receive bonuses sized with a full hierarchy of length 5.
In PtQ, 5 opt out, and the hierarchy is of full length 5.

Why is PtQ optimal?

Contracts robustly implementing all-high-quality

- need not be uniformly PO or PtQ across experts and undertaking profiles, $(i, \mathbf{a}_{-i})$,
- but at each $(i, \mathbf{a}_{-i})$, $\mathbf{b}$ must either *pool* or *separate* i's type.

The design is a local choice between PO and PtQ.

Key spillover: At each $(i, \mathbf{a}_{-i})$, switching from PO to PtQ

- is *free* for the principal,
- makes undertaking *informative* about high quality,
- *lowers* the bonuses required by experts resolved earlier than i in the elimination order.

Optimal contract

The logic extends to n experts with heterogeneous c_i and str. inc. and str. supermodular $P(Q)$.

The problem collapses to finding an optimal order of eliminations:

$$\sigma^* \in \operatorname*{argmin}_{\sigma \in \mathrm{Perm}(N)} C^{\mathrm{PtQ}}(\sigma, \mathbf{p})$$

Proposition (Optimal UI contract)

The optimal contract among contracts that robustly implement all-high-quality is a PtQ contract for all $(i, \mathbf{a}_{-i})$, for some $\sigma \in \mathrm{Perm}(N)$.

In general, σ^* admits no closed-form characterization.

▸ Proof sketch

In O-ring production, the optimal order ranks experts who are cheap / critical / likely-already-satisfactory earlier.

Final remarks

Result.

The optimal contract pays experts to opt out while robustly inducing repair of every faulty part.

Mechanism.

The incentive scheme makes an already observable choice informative.

Team spillover.

Making one expert's choice informative lowers the bonuses needed to incentivize other experts. The value of separation extends beyond the expert being separated.

Related literature

Contracting with externalities

- Segal (2003): divide-and-conquer under multilateral externalities
- Bernstein & Winter (2012): heterogeneous participation externalities

Team moral hazard with complementarities — unique implementation under action uncertainty

- Winter (2004): hierarchical contracts
- Halac, Lipnowski & Rappoport (2021): principal-designed uncertainty

Robust implementation — robust to beliefs about types

- Bergemann & Morris (2017); Dekel, Fudenberg & Morris (2007)

Mechanism design with evidence — exogenous evidence structure

- Green & Laffont (1986); Bull & Watson (2007)

This paper: unique implementation robust to *both* type and action uncertainty,
with *endogenous* evidence created through contract design.

Appendix

Belief-Free Rationalizability

Definition (Belief-free rationalizable actions)

Initialize: $B^{\mathbf{b}}_{i,0}(\theta_i) = \mathcal{A}_i = \{(0,S), (1,S), (1,W)\}$ for all $i \in N$, $\theta_i \in \Theta$.

Iterate: $(a_i, e_i) \in B^{\mathbf{b}}_{i,k}(\theta_i)$ survives in $B^{\mathbf{b}}_{i,k+1}(\theta_i)$ iff there exists an arbitrary conjecture $\nu \in \Delta(\Theta_{-i} \times \mathcal{A}_{-i})$ such that:

(i) $\nu(\boldsymbol{\theta}_{-i}, \mathbf{a}_{-i}, \mathbf{e}_{-i}) > 0 \Rightarrow (a_j, e_j) \in B^{\mathbf{b}}_{j,k}(\theta_j)$ for all $j \neq i$;

(ii) $(a_i, e_i) \in \arg\max_{(a_i', e_i') \in \mathcal{A}_i} \sum u_i^{\mathbf{b}}(\theta_i, a_i', e_i' \mid \boldsymbol{\theta}_{-i}, \mathbf{a}_{-i}, \mathbf{e}_{-i})\, \nu(\cdot)$.

Limit: $B_i(\theta_i \mid \mathbf{b}) = \bigcap_{k \geq 0} B^{\mathbf{b}}_{i,k}(\theta_i)$.

Back

Optimal contract: Proof sketch

Idea: Start from any contract.
We can always find a PtQ contract that is cheaper for each realization.

Start from any robust all-high-quality contract $\mathbf{b}$:

1. Characterize: Let $\tau(i)$=the elimination round for Expert i and $S_i(\mathbf{a}_{-i})$=the assured sets.
2. Fill in: Choose a ranking σ refining (breaking ties in) τ.
   Extend bonuses to all off-path profiles (unreachable) at critical values $\Rightarrow$ same cost.
3. Reduce: Replace bonuses with critical values along σ $\Rightarrow$ weakly lower cost.
4. Switch to PtQ: expert by expert, PO-form $\rightarrow$ PtQ-form
   Each switch preserves UI and enlarges predecessors' assured sets $\Rightarrow$ weakly lower cost.
5. Optimize: $C(\mathbf{b}) \geq C^{PtQ}(\sigma, \mathbf{p}) \geq \min_{\sigma'} C^{PtQ}(\sigma', \mathbf{p})$.

Back

General setting

Environment

- n experts, heterogeneous type dist. $p_i = \Pr(\theta_i = H) \in (0,1)$ and effort costs $c_i > 0$
- Success probability $P(Q)$ is *strictly increasing* and *strictly supermodular* in the high-quality set Q:

$$P(Q \cup \{i\}) - P(Q) \;<\; P(Q' \cup \{i\}) - P(Q') \qquad \forall\, i,\; Q \subsetneq Q' \subseteq N \setminus \{i\}$$

Robust implementation (RI): **b** makes all-high-quality the *unique* BFR outcome.

(up to a vanishing tie-breaking perturbation, since the RI set is typically open)

Optimal RI contract

$$\min_{\mathbf{b} \in \mathrm{RI}} \; P(N)\, \mathbf{E_p}\left[\sum_{i \in N} b_i\big(\mathbf{a}(\theta)\big)\right]$$

PtQ: General characterization

$$b_i^* = \frac{c_i}{P(W_{-i} \cup \{i\} \cup R_{<i}^{\sigma}) - P(W_{-i} \cup R_{<i}^{\sigma})}$$

where $W(\mathbf{a}) = \{j \in N : a_j = 1\}$ is the set of participants who are guaranteed to work and hence produce q_H, and
$R_{<i}^{\sigma}$ is the set of resolved experts before Expert i.

Perturbations: Set $b_i(1, \mathbf{a}_{-i}) = b_i^* + \varepsilon_{i,1}$ and $b_i(0, \mathbf{a}_{-i}) = b_i^* + \varepsilon_{i,0}$.

$$1 < \frac{\varepsilon_{i,0}}{\varepsilon_{i,1}} < \frac{P(W_{-i} \cup \{i\} \cup R_{<i}^{\sigma})}{P(W_{-i} \cup R_{<i}^{\sigma})} \qquad \forall\, \mathbf{a}_{-i}$$

must hold to break ties to induce the desired action for each type.

By strict monotonicity, $\frac{P(W_{-i} \cup \{i\} \cup R_{<i}^{\sigma})}{P(W_{-i} \cup R_{<i}^{\sigma})} > 1$ for all $\mathbf{a}_{-i}$. Such perturbations always exist.

Special case: O-ring production

O-ring technology: $P(Q) = \prod_{i \notin Q} \lambda_i, \;\; 0 < \lambda_i < 1$
Every expert is a "generalist": i's marginal contribution $\propto P(Q_{-i})$.

Closed form: Each later-resolved j inflates every predecessor's bonus by a loading $\kappa_j := (1 - p_j) + \dfrac{p_j}{\lambda_j} > 1$:

$$b^*_{\sigma(i)} = \underbrace{\frac{c_i}{1 - \lambda_i}}_{\text{base bonus}} \prod_{\sigma(j) > \sigma(i)} \kappa_j$$

Proposition (Optimal ranking under O-ring)

The optimal contract ranks experts by increasing priority index $I_i = \dfrac{c_i \, \lambda_i}{p_i (1 - \lambda_i)^2}$ *and offers PtQ for each* $(i, \mathbf{a}_{-i})$.

Rank early: *cheap* (low c_i), *critical* (low λ_i), *likely-already-good* (high p_i).
Symmetric λ: $I_i \propto c_i / p_i$. Symmetric λ & c: decreasing p_i (most-likely-good first).

▸ Proof

Fragile technology

Intervention without care *degrades* quality:

$$q(\theta_i, a_i, e_i) = \begin{cases} q_H & \text{if } \theta_i = H, a_i = 0 \\ q_H & \text{if } a_i = 1 \text{ and } e_i = W \\ q_L & \text{otherwise} \end{cases}$$

Now two contracts can separate types:

Contract 1 (PtQ): $b_i(0, \mathbf{a}_{-i}) > b_i(1, \mathbf{a}_{-i})$ (same as before)

- Participants = assured high quality. Better when defects are common.

Contract 2 (PtP): $b_i(1, \mathbf{a}_{-i}) > b_i(0, \mathbf{a}_{-i}) > b_i(1, \mathbf{a}_{-i}) - c_i$

- Non-participants = assured high quality. Better when parts are mostly stable.
- After type separation, hierarchical wage sizing for participants

Optimal: assign PtQ when $p_i \leq 1/2$, PtP when $p_i > 1/2$.

O-Ring characterization proof

Under $P(Q) = \prod_{i \notin Q} \lambda_i$, expected cost under ranking σ:

$$C_{\mathbf{p}}^{P \text{ to } Q}(\sigma) = \sum_{k=1}^{n} \frac{c_{i_k}}{1 - \lambda_{i_k}} \prod_{\ell > k} \kappa_{i_\ell}, \qquad \kappa_i := (1 - p_i) + \frac{p_i}{\lambda_i}$$

Adjacent swap condition: rank i before j iff $\dfrac{c_i/(1-\lambda_i)}{\kappa_i - 1} \leq \dfrac{c_j/(1-\lambda_j)}{\kappa_j - 1}$.

$\Longrightarrow$ Optimal ranking: increasing $I_i = \dfrac{c_i\, \lambda_i}{p_i (1-\lambda_i)^2}$.

Back